



Sotos Syndrome

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Summary

Clinical characteristics

Sotos syndrome is characterized by a distinctive facial appearance (broad, prominent forehead with a dolichocephalic head shape, sparse frontotemporal hair, downslanting palpebral fissures, malar flushing, long and narrow face, tall chin); learning disability (early developmental delay, mild-to-severe intellectual impairment); and overgrowth (height and/or head circumference ≥ 2 SD above the mean). These three clinical features (distinctive facial features, learning disability, and overgrowth) are considered the cardinal features of Sotos syndrome. Major features of Sotos syndrome include behavioral findings (most notably autism spectrum disorder), advanced bone age, cardiac anomalies, cranial MRI/CT abnormalities, joint hyperlaxity with or without pes planus, maternal preeclampsia, neonatal complications, renal anomalies, scoliosis, and seizures.

Diagnosis/testing

The diagnosis of Sotos syndrome is established in a proband with a heterozygous *NSD1* pathogenic variant or a deletion encompassing *NSD1* identified by molecular genetic testing.

Management

Treatment of manifestations: Referral to appropriate specialists for management of learning disability / speech delays, behavioral findings, cardiac abnormalities, renal anomalies, scoliosis, and seizures; intervention is not recommended if the brain MRI shows ventricular dilatation without increased intracranial pressure.

Surveillance: Regular review by a general pediatrician for younger children, individuals with many medical complications, and families requiring more support than average; less frequent review of older children / teenagers and those individuals without many medical complications.

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Genetic counseling

Sotos syndrome is inherited in an autosomal dominant manner. About 5% of individuals diagnosed with Sotos syndrome have an affected parent; approximately 95% of individuals have the disorder as the result of a *de novo* genetic alteration. Each child of an individual with Sotos syndrome has a 50% chance of inheriting the causative genetic alteration. Once the Sotos syndrome-related genetic alteration has been identified in an affected family member, prenatal and preimplantation genetic testing are possible. Phenotypic expression can vary from one generation to the next; thus, it is not possible to accurately predict phenotype based on the prenatal finding of a Sotos syndrome-related genetic alteration.

Diagnosis

No consensus clinical diagnostic criteria Sotos syndrome have been published.

Suggestive Findings

Sotos syndrome **should be suspected/considered** in probands with the following features.

Characteristic facial appearance (most easily recognizable between ages 1 and 6 years):

- Broad, prominent forehead with a dolichocephalic head shape
- Sparse frontotemporal hair
- Downslanting palpebral fissures
- Malar flushing
- Long narrow face (particularly bitemporal narrowing)
- Tall chin

Note: Facial shape is retained into adulthood. However, with time the chin becomes broader (squarer in shape).

Learning disability

- Early developmental delay
- Mild-to-severe intellectual impairment

Overgrowth

- Tall stature. Height ≥ 2 standard deviations (SD) above the mean (i.e., ~98th centile). Note: Height may normalize in adulthood.
- Macrocephaly. Head circumference ≥ 2 SD above the mean (i.e., ~98th centile), usually present at all ages. Congenital macrocephaly is observed in most individuals. Head growth velocity tends to increase during the first year of life, and macrocephaly persists into adulthood [Cole & Hughes 1994, Foster et al 2019].

Note: Based on the analysis of more than 266 individuals with an *NSD1* pathogenic variant, the three cardinal features (facial appearance, learning disability, and overgrowth) were shown to occur in at least 90% of affected individuals [Tatton-Brown et al 2005b].

Establishing the Diagnosis

The diagnosis of Sotos syndrome **is established** in a proband with a heterozygous pathogenic (or likely pathogenic) variant in *NSD1* or a deletion encompassing *NSD1* identified on molecular genetic testing (see Table 1).

Note: (1) Per ACMG/AMP variant interpretation guidelines, the terms "pathogenic variant" and "likely pathogenic variant" are synonymous in a clinical setting, meaning that both are considered diagnostic and can

be used for clinical decision making [Richards et al 2015]. Reference to "pathogenic variants" in this *GeneReview* is understood to include likely pathogenic variants. (2) Identification of a heterozygous *NSD1* variant of uncertain significance does not establish or rule out the diagnosis.

Molecular genetic testing approaches can include a combination of **gene-targeted testing** (single gene testing, multigene panel) and **comprehensive genomic testing** (exome sequencing, genome sequencing). Gene-targeted testing requires that the clinician determine which gene(s) are likely involved (see Option 1), whereas comprehensive genomic testing does not (see Option 2).

Option 1

When the phenotypic and laboratory findings suggest the diagnosis of Sotos syndrome, molecular genetic testing approaches can include **single-gene testing** or use of a **multigene panel**:

- **Single-gene testing.** Sequence analysis of *NSD1* is performed first to detect missense, nonsense, and splice site variants and small intragenic deletions/insertions. Note: Depending on the sequencing method used, single-exon, multiexon, or whole-gene deletions/duplications may not be detected. If no variant is detected by the sequencing method used, the next step is to perform gene-targeted deletion/duplication analysis to detect exon and whole-gene deletions or duplications.
- **Chromosomal microarray analysis (CMA)** uses oligonucleotide or SNP arrays to detect genome-wide large deletions/duplications (including *NSD1*) that cannot be detected by sequence analysis. Many individuals with developmental delay will have had CMA as part of their evaluation, sometimes prior to the consideration of Sotos syndrome as a possibility. CMA designs in current clinical use target the *NSD1* region. Gene-targeted deletion/duplication assays may have higher resolution than CMA; however, testing is unlikely to detect a deletion or duplication in cases where sequence analysis and CMA of *NSD1* was not diagnostic.

For an introduction to CMA click [here](#). More detailed information for clinicians ordering genetic tests can be found [here](#).

- **A multigene panel** that includes *NSD1* and other genes of interest (see Differential Diagnosis) is most likely to identify the genetic cause of the condition while limiting identification of pathogenic variants and variants of uncertain significance in genes that do not explain the underlying phenotype. Note: (1) The genes included in the panel and the diagnostic sensitivity of the testing used for each gene vary by laboratory and are likely to change over time. (2) Some multigene panels may include genes not associated with the condition discussed in this *GeneReview*. (3) In some laboratories, panel options may include a custom laboratory-designed panel and/or custom phenotype-focused exome analysis that includes genes specified by the clinician. (4) Methods used in a panel may include sequence analysis, deletion/duplication analysis, and/or other non-sequencing-based tests.

For an introduction to multigene panels click [here](#). More detailed information for clinicians ordering genetic tests can be found [here](#).

Option 2

When the diagnosis of Sotos syndrome is not considered because an individual has atypical phenotypic features, **comprehensive genomic testing** (which does not require the clinician to determine which gene[s] are likely involved) is the best option. **Exome sequencing** is the most commonly used genomic testing method; **genome sequencing** is also possible. To date, the majority of *NSD1* pathogenic variants reported (e.g., missense, nonsense) are within the coding region and are likely to be identified on exome sequencing. Literature is emerging regarding the observation of novel splice site and intronic *NSD1* variants [Minatogawa et al 2022, Villate et al 2022, Testa et al 2023]. Further pathogenic variants are likely to be identified in accompaniment with increasing use of RNA sequencing technologies.

For an introduction to comprehensive genomic testing click [here](#). More detailed information for clinicians ordering genomic testing can be found [here](#).

Table 1. Molecular Genetic Testing Used in Sotos Syndrome

Gene ¹	Method	Proportion of Pathogenic Variants ² Identified by Method
<i>NSD1</i>	Sequence analysis ³	45%-80% ⁴
	CMA ⁵	15%-50% ⁶
	Gene-targeted deletion/duplication analysis ⁷	20%-55% ^{7, 8}

- 1. See Table A. Genes and Databases for chromosome locus and protein.
- 2. See Molecular Genetics for information on variants detected in this gene.
- 3. Sequence analysis detects variants that are benign, likely benign, of uncertain significance, likely pathogenic, or pathogenic. Variants may include missense, nonsense, and splice site variants and small intragenic deletions/insertions; typically, exon or whole-gene deletions/duplications are not detected. For issues to consider in interpretation of sequence analysis results, click [here](#).
- 4. Data derived from the subscription-based professional view of Human Gene Mutation Database [Stenson et al 2020]
- 5. Chromosomal microarray analysis (CMA) uses oligonucleotide or SNP arrays to detect genome-wide large deletions/duplications (including *NSD1*) that cannot be detected by sequence analysis. The ability to determine the size of the deletion/duplication depends on the type of microarray used and the density of probes in the 5q35 region. Typically, single-exon deletions or duplications are below the resolution of CMA detection. CMA designs in current clinical use target the 5q35 region and *NSD1*.
- 6. A recurrent 1.9-Mb 5q35 microdeletion encompassing *NSD1* has been reported in most Japanese and some non-Japanese individuals with Sotos syndrome [Kurotaki et al 2003, Tatton-Brown et al 2005a, Tatton-Brown et al 2005b, Visser et al 2005]. Further data about population distribution of this microdeletion are not yet available. Most microdeletions are generated by nonallelic homologous recombination between flanking low-copy repeats [Kurotaki et al 2003, Tatton-Brown et al 2005a, Visser et al 2005]. Many of these recurrent deletions have the same breakpoints, and a specific chromatin structure may increase recurrent crossover events and predispose to recombination hot spots at 5q35 [Visser et al 2005]. The incidence of the common 1.9-Mb deletion varies by population [Kurotaki et al 2003, Tatton-Brown et al 2005a, Tatton-Brown et al 2005b, Visser et al 2005].
- 7. Gene-targeted deletion/duplication analysis detects intragenic deletions or duplications. Methods used may include a range of techniques such as quantitative PCR, long-range PCR, multiplex ligation-dependent probe amplification (MLPA), and a gene-targeted microarray designed to detect single-exon deletions or duplications. Gene-targeted deletion/duplication testing will detect deletions ranging from a single exon to the whole gene; however, breakpoints of large deletions and/or deletion of adjacent genes (e.g., the 1.9-Mb deletion) may not be detected by these methods. Exome and genome sequencing may be able to detect deletions/duplications using breakpoint detection or read depth; however, sensitivity can be lower than gene-targeted deletion/duplication analysis.
- 8. Gene-targeted deletion/duplication analysis may detect an additional 1%-5% of partial-gene deletions.

Epigenetic signature analysis / methylation array. A distinctive epigenetic signature (disorder-specific genome-wide changes in DNA methylation profiles) in peripheral blood leukocytes has been identified in individuals with Sotos syndrome [Aref-Eshghi et al 2019, Aref-Eshghi et al 2020, Levy et al 2021]. The Sotos syndrome / *NSD1* epigenetic signature is known to be highly sensitive [Husson et al 2024]. Epigenetic signature analysis of a peripheral blood sample or DNA banked from a blood sample can therefore be considered to clarify the diagnosis in individuals with: (1) suggestive clinical findings of Sotos syndrome but in whom no pathogenic or likely pathogenic variant in *NSD1* has been identified via sequence analysis or CMA; or (2) suggestive clinical findings of Sotos syndrome and a *NSD1* variant of uncertain clinical significance identified by molecular genetic testing. For an introduction to epigenetic signature analysis click [here](#).

Clinical Characteristics

Clinical Description

To date, more than 900 individuals have been identified with a pathogenic variant in *NSD1* or deletion encompassing *NSD1*. The largest study to date reviewed 266 persons with *NSD1* abnormalities [Tatton-Brown et al 2005b]. Based on this review, the clinical features of Sotos syndrome were classified as cardinal features (occurring in ≥90% of affected individuals), major features (occurring in 15%-89%), and associated features

(occurring in $\geq 2\%$ and $< 15\%$ of persons) [Tatton-Brown et al 2005b]. While a few single case reports have been published since this time, no robust additional associations have been delineated. The following description of the phenotypic features associated with this condition is based on these reports.

Table 2. Sotos Syndrome: Frequency of Select Features

Class	Features	Comment
Cardinal features (present in $\geq 90\%$ of persons w/Sotos syndrome)	Characteristic facial appearance	Dolichocephaly, prominent forehead, sparse frontotemporal hair, downslanting palpebral fissures, malar flushing
	Learning disability	Variable
	Overgrowth (height &/or head circumference ≥ 2 SD above mean)	
Major features (present in 15%-89% of persons w/Sotos syndrome)	Behavioral findings	Most notably ASD
	Advanced bone age	
	Cardiac anomalies	Variable (incl PDA, septal defects, &/or more complex anomalies)
	Brain MRI/CT abnormalities	Most commonly ventriculomegaly
	Joint hyperlaxity w/ or w/o pes planus	
	Maternal preeclampsia	
	Neonatal complications	Hypotonia, jaundice, poor feeding
	Renal anomalies	Most commonly VUR
	Scoliosis	
	Seizures	

Based on Tatton-Brown et al [2005b]

ASD = autism spectrum disorder; PDA = patent ductus arteriosus; SD = standard deviations; VUR = vesicoureteral reflux

Cardinal Features

Characteristic facial appearance. The facial gestalt is the most specific diagnostic criterion for Sotos syndrome, and the one most open to observer error due to inexperience. The facial gestalt of Sotos syndrome is evident at birth but becomes most recognizable between ages one and six years. The head is dolichocephalic and the forehead broad and prominent. Often the hair in the frontotemporal region is sparse. The palpebral fissures are usually downslanting. Malar flushing may be present. In childhood the jaw is narrow with a tall chin; in adulthood the chin broadens [Allanson & Cole 1996, Tatton-Brown & Rahman 2004]. In older children and adults, the facial features, although still typical, can be more subtle [Allanson & Cole 1996, Tatton-Brown et al 2005b].

Learning disability. Delay of early developmental milestones is very common, and motor skills may appear particularly delayed because of a child's large size, hypotonia, and poor coordination. Most individuals with Sotos syndrome have some degree of intellectual impairment. The spectrum is broad and ranges from a mild learning disability (affected individuals would be expected to live independently and have their own families) to a severe learning disability (affected individuals would be unlikely to live independently as adults). The level of intellectual impairment generally remains stable throughout life [Tatton-Brown et al 2005b, Foster et al 2019]. The learning disability in children with Sotos syndrome is characterized by relative strength in verbal ability and

visuospatial memory but relative weakness in nonverbal reasoning ability and quantitative reasoning [Lane et al 2019].

Overgrowth. Sotos syndrome is associated with overgrowth of prenatal onset. Delivery is typically at term. The average birth length approximates to the 98th centile and the average birth head circumference is between the 91st and 98th centiles. Average birth weight is within the normal range (50th-91st centile).

Before age ten years, affected children often demonstrate rapid linear growth. They are often described as being considerably taller than their peers. Approximately 90% of children have a height and/or head circumference at least two standard deviations (SD) above the mean [Tatton-Brown et al 2005b]. However, growth is also influenced by parental heights, and some individuals do not have growth parameters above the 98th centile [Cole & Hughes 1994, Tatton-Brown et al 2005b].

Height may normalize in adulthood, but macrocephaly is usually present at all ages [Agwu et al 1999, Foster et al 2019]. Data on final adult height are scarce; however, in both men and women, the range of final adult height is broad [Agwu et al 1999, Foster et al 2019].

Auxologic data shows that individuals with an *NSD1* pathogenic variant have an increased arm span-to-height ratio, decreased sitting-to-standing height ratio, and increased hand length. These data suggest that the increased height in Sotos syndrome is predominantly the result of an increase in limb length [Agwu et al 1999, de Boer et al 2005].

Major Features

Behavioral findings. A wide range of behavioral findings are common at all ages. Autism spectrum disorder, phobias, sleep disturbances, hyperactivity, and aggression have been described [Sheth et al 2015; Lane et al 2017; Foster et al 2019; Lane et al 2019; Frattale et al 2024; K Tatton-Brown, unpublished data]. Often difficulty with peer group relationships is precipitated by large size, naiveté, and lack of awareness of social cues [Finegan et al 1994]. These observations were confirmed in a study of individuals with a clinical diagnosis of Sotos syndrome (some with and some without a known *NSD1* pathogenic variant); it was additionally noted that attention-deficit/hyperactivity disorder is not common among individuals with Sotos syndrome [de Boer et al 2006].

Advanced bone age. Bone age often reflects the accelerated growth velocity and is advanced in 75%-80% of prepubertal children. However, bone age interpretation is influenced by the "threshold" used, the method of assessment, subjective interpretative error, and the age at which the assessment is made.

Cardiac anomalies. About 15%-40% of individuals have cardiac anomalies that range in severity from predominantly single, often self-limiting anomalies (including patent ductus arteriosus, atrial septal defect, and ventricular septal defect) to more severe, complex cardiac abnormalities.[Calcagni et al 2024]. Six unrelated individuals with Sotos syndrome have been shown to have left ventricular non-compaction [Martinez et al 2011, Saccucci et al 2011, Nakamura et al 2014, Calcagni et al 2024]. Aortic dilatation has been reported in 16 individuals to date [Hood et al 2016, Foster et al 2019, Pezzani et al 2020, Calcagni et al 2024].

Brain MRI/CT abnormalities are identified in the majority of individuals with an *NSD1* pathogenic variant. Ventricular dilatation (particularly in the trigone region) is most frequently identified; other abnormalities include midline changes (hypoplasia or agenesis of the corpus callosum, mega cisterna magna, cavum septum pellucidum), cerebral atrophy, and small cerebellar vermis [Waggoner et al 2005].

Joint hyperlaxity / pes planus. Joint laxity is reported in at least 20% of individuals with Sotos syndrome.

Maternal preeclampsia occurs in about 15% of pregnancies of children with Sotos syndrome.

Neonatal complications. Neonates may have jaundice (~65%), hypotonia (~75%), and poor feeding (~70%). These complications tend to resolve spontaneously, but in a small minority intervention is required.

Renal anomalies. About 15% of individuals with an *NSD1* pathogenic variant have a renal anomaly; vesicoureteral reflux is the most common. Some individuals may have quiescent vesicoureteral reflux and may present in adulthood with renal impairment.

Scoliosis. Present in about 30% of affected individuals, scoliosis is rarely severe enough to require bracing or surgery.

Seizures. Approximately 25% of individuals with Sotos syndrome develop non-febrile seizures at some point in their lives and some require ongoing therapy. Absence, tonic-clonic, myoclonic, and partial complex seizures have all been reported.

Associated Features

Tumors occur in approximately 3% of persons with Sotos syndrome. Several types have been observed including sacrococcygeal teratoma (4 individuals), neuroblastoma (2 individuals), and single case reports of hepatoblastoma, presacral and intracranial ganglioma, acute lymphoblastic leukemia, small-cell lung cancer, astrocytoma, and pineoblastoma [Hersh et al 1992, Deardorff et al 2004, Tatton-Brown & Rahman 2004, Saugier-Verber et al 2007, Theodoulou et al 2015, Grand et al 2019, Yue et al 2024].

De Boer and colleagues have characterized and reviewed these problems and compared persons with Sotos syndrome who have *NSD1* pathogenic variants to those who do not [de Boer et al 2006]. Given that many of these are single case reports, it is unclear whether these tumors are specifically associated with Sotos syndrome or coincidental findings.

Various other clinical features have been associated with Sotos syndrome. Some associated features, such as constipation and hearing problems caused by chronic otitis media, are common. The following features are seen in $\geq 2\%$ and $< 15\%$ of individuals with Sotos syndrome [Tatton-Brown et al 2005b]:

- Ocular. Astigmatism, cataracts, myopia, strabismus, nystagmus, hypermetropia
- Auricular. Cholesteatoma, sensorineural and conductive hearing loss
- Gastrointestinal. Constipation, gastroesophageal reflux, Hirschsprung disease
- Genitourinary. Cryptorchidism, hypospadias, hydrocele, phimosis
- Musculoskeletal. Pectus excavatum, talipes equinovarus, vertebral anomalies, contractures, 2-3 toe syndactyly
- Skin. Hemangiomas, hyper- or hypopigmentation
- Craniofacial. Craniosynostosis
- Endocrine. Hypothyroidism, neonatal hypoglycemia
- Renal. Nephrocalcinosis
- Lungs. Subpleural blebs
- Hemihypertrophy
- Umbilical and/or inguinal hernias
- Nail and tooth anomalies. Hypoplastic nails, hypodontia

Genotype-Phenotype Correlations

Through the evaluation of 234 individuals with Sotos syndrome with an *NSD1* abnormality, it has been shown that individuals with a 5q35 microdeletion have less overgrowth and more severe learning disability than individuals with an *NSD1* intragenic pathogenic variant overall [Tatton-Brown et al 2005b].

Genotype-phenotype correlations for intragenic pathogenic variants and 5q35 microdeletions are not evident for other clinical features associated with Sotos syndrome (i.e., cardiac abnormalities, renal anomalies, seizures, scoliosis), nor were correlations observed between the type of intragenic pathogenic variant (missense vs

truncating) and phenotype or between position of pathogenic variant (5' vs 3' UTR) and phenotype [Tatton-Brown et al 2005b]. Recent studies have replicated these findings [Choi et al 2024].

Penetrance

To date, no unaffected parent or sib with an *NSD1* pathogenic variant or deletion encompassing *NSD1* has been reported [Douglas et al 2003, Rio et al 2003, Türkmen et al 2003, Tatton-Brown et al 2005b]. Thus, Sotos syndrome appears to be a fully penetrant condition.

Of note, expressivity is highly variable. Individuals with the same genetic alteration, even within the same family, can be affected differently [Tatton-Brown et al 2005b, Donnelly et al 2011].

Nomenclature

Sotos syndrome has previously been referred to as cerebral gigantism. This term is now outdated and should no longer be used. Sotos syndrome is included under the umbrella term of overgrowth-intellectual disability (OGID) syndromes [Tatton-Brown et al 2017].

Prevalence

Sotos syndrome is estimated to occur in 1:14,000 live births [K Tatton-Brown, TRP Cole, N Rahman, unpublished data].

Genetically Related (Allelic) Disorders

No phenotypes other than those discussed in this *GeneReview* are known to be associated with germline pathogenic variants in *NSD1*. *NSD1* abnormalities have a high specificity and sensitivity for Sotos syndrome. Molecular genetic testing failed to identify pathogenic variants in *NSD1* in more than 500 individuals with clinical diagnoses other than Sotos syndrome, including Marshall-Smith syndrome, nonspecific overgrowth, and individuals with macrocephaly in association with autism spectrum disorders [Türkmen et al 2003, Tatton-Brown et al 2005b, Waggoner et al 2005, Buxbaum et al 2007].

Contiguous gene duplications involving *NSD1* are associated with "reverse Sotos syndrome," a phenotype characterized by short stature, microcephaly, developmental delay, characteristic facial features, and variable congenital abnormalities [Takkar et al 2022, van der Lugt et al 2023].

Differential Diagnosis

Overgrowth conditions that may be confused with Sotos syndrome are summarized in Table 3.

Table 3. Overgrowth Conditions to Consider in the Differential Diagnosis of Sotos Syndrome

Gene(s) / Genetic Mechanism	Disorder	MOI	Clinical Features of Disorder	
			Overlapping w/Sotos syndrome	Distinguishing from Sotos syndrome
Key differential diagnoses				

Table 3. continued from previous page.

Gene(s) / Genetic Mechanism	Disorder	MOI	Clinical Features of Disorder	
			Overlapping w/Sotos syndrome	Distinguishing from Sotos syndrome
<i>EED</i>	<i>EED</i> -related overgrowth (Cohen-Gibson syndrome)	AD	• Typical but subtle facial appearance, esp in early childhood • ↑ height, macrocephaly, scoliosis, ligamentous laxity • Frequently hypotonic at birth (may present w/ mixed central hypotonia / peripheral hypertonia)	• The face is subtly different, w/ hypertelorism, round face, & "stuck-on" chin. • Other features that do not overlap w/Sotos syndrome are most similar to <i>EZH2</i>-related Weaver syndrome.
<i>EZH2</i>	<i>EZH2</i> -related Weaver syndrome ^{1, 2} (See <i>EZH2</i> -Related Overgrowth.)	AD	• Pre- & postnatal overgrowth • Variable ID • Similar (but distinctive) facial appearance • Advanced bone age • Scoliosis • Joint hypermobility	• Sotos syndrome & <i>EZH2</i>-related overgrowth are most easily distinguishable at ages 1-3 yrs. • Round face shape w/ocular hypertelorism • Prognathism is not observed, but chin appears "stuck on," frequently w/horizontal crease between chin & lower lip. • Assoc joint problems (e.g., camptodactyly & contractures) • Infants often have hoarse, low-pitched cry.
<i>SUZ12</i>	<i>SUZ12</i> -related overgrowth syndrome (Imagawa-Matsumoto syndrome) (OMIM 618786)	AD	• Typical but subtle facial appearance, esp in early childhood • ↑ height, macrocephaly, scoliosis, ligamentous laxity • Frequently hypotonic at birth (may present w/ mixed central hypotonia / peripheral hypertonia)	Features that do not overlap w/Sotos syndrome are most similar to <i>EZH2</i> -related Weaver syndrome.
Other disorders of interest				
Abnormal regulation of gene transcription in 2 imprinted domains at 11p15.5 ³	Beckwith-Wiedemann syndrome (BWS)	AD	Frequently, weight &/or height are ≥2 SD at birth.	• BWS should be clinically distinguishable from Sotos syndrome (molecular testing is indicated in persons w/ clinical overlap). • Macroglossia • Anterior earlobe creases / helical pits • Omphalocele • Visceromegaly • ↑ risk of embryonal tumors, esp Wilms tumor

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Gene(s) / Genetic Mechanism	Disorder	MOI	Clinical Features of Disorder	
			Overlapping w/Sotos syndrome	Distinguishing from Sotos syndrome
<i>DNMT3A</i>	Tatton-Brown-Rahman syndrome (<i>DNMT3A</i> -related overgrowth syndrome)	AD	- Overgrowth - Variable ID - Joint hypermobility - Scoliosis	- Round face w/coarse features - Prominent central incisors, narrow palpebral fissures, thick, low-set eyebrows - Tendency toward overweight & obesity
<i>FMR1</i>	Fragile X syndrome (See FMR1-Related Disorders .)	XL	- Macrocephaly - ID - Typical but subtle facial appearance may overlap w/Sotos syndrome: dolichocephalic head shape, prominent jaw & forehead.	- Predominantly affects males - Affected males often have large ears, which is not common in Sotos syndrome. - Macro-orchidism after puberty
<i>GPC3</i> PV or whole-gene deletion of <i>GPC3</i>	Simpson-Golabi-Behmel syndrome type 1	XL	- Pre- & postnatal overgrowth - Variable ID	- Predominantly affects males - Polydactyly - Supernumerary nipples - Diastasis recti - Pectus excavatum - Facial gestalt differs
<i>NFIX</i>	Malan syndrome	AD	- Characteristic facial appearance w/ dolichocephaly, prominent forehead, & downslanting palpebral fissures - Scoliosis, ↑ height (>2 SD) in childhood - Assoc w/variable ID	- Ophthalmic abnormalities are common in Malan syndrome & less common in Sotos syndrome. ↑ height is less common in older children & adults.
<i>PTCH</i> <i>SUFU</i>	Nevoid basal cell carcinoma syndrome (Gorlin syndrome)	AD	- Recognizable appearance w/macrocephaly, bossing of forehead, & coarse facial features (in ~60%) - Head circumference ↑ to >98th centile until age 10-18 mos	- Development of multiple jaw keratocysts, frequently beginning in 2nd decade - Basal cell carcinomas usually from 3rd decade onward - Skeletal anomalies (e.g., bifid ribs or wedge-shaped vertebrae) - Absence, usually, of DD

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Gene(s) / Genetic Mechanism	Disorder	MOI	Clinical Features of Disorder	
			Overlapping w/Sotos syndrome	Distinguishing from Sotos syndrome
<i>PTEN</i>	Bannayan-Riley-Ruvalcaba syndrome (See PTEN Hamartoma Tumor Syndrome .)	AD	- Macrocephaly - Somewhat similar facial gestalt - May be assoc w/ASD	- Vascular malformations - Hamartomatous polyps of distal ileum & colon - Pigmented macules on shaft of penis - Lipomas ↑ risk of thyroid & breast cancer

AD = autosomal dominant; AR = autosomal recessive; ASD = autistic spectrum disorder; DD = developmental delay; ID = intellectual disability; MOI = mode of inheritance; PV = pathogenic variant; XL = X-linked

1. Given the overlap in findings between Sotos syndrome, *EZH2*-related Weaver syndrome, *EED*-related overgrowth, and *SUZ12*-related overgrowth, in probands with features suggestive of *EZH2*-related Weaver syndrome, the recommended testing order is *NSD1*, followed by *EZH2*, followed by *EED*, followed by *SUZ12*.

2. In some individuals thought to have *EZH2*-related Weaver syndrome but in whom *NSD1* pathogenic variants were identified, the clinical phenotype evolved to be consistent with Sotos syndrome [Douglas et al 2003, Rio et al 2003, Tatton-Brown et al 2005b].

3. Beckwith-Wiedemann syndrome (BWS) without multilocus imprinting disturbances is associated with abnormal expression of imprinted genes in the BWS critical region. Abnormal expression of imprinted genes can be caused by an epigenetic or genomic alteration leading to an abnormal methylation pattern at 11p15.5, a copy number variant of chromosome 11p15.5, or a heterozygous maternally inherited *CDKN1C* pathogenic variant. Reliable recurrence risk assessment requires identification of the genetic mechanism in the proband that underlies the abnormal expression of imprinted genes in the BWS critical region. While most families have a recurrence risk of less than 1%, certain underlying genetic mechanisms involve a recurrence risk as high as 50%.

Other conditions to consider in the differential diagnosis of Sotos syndrome include the following:

- **Chromosomal abnormalities.** A Sotos syndrome-like phenotype has been associated with 4p duplications, mosaic 20p trisomy [Faivre et al 2000], and [Phelan-McDermid syndrome](#) (22q13.3 deletion syndrome).
- **Nonspecific overgrowth.** Many individuals with overgrowth do not fulfill the diagnostic criteria for any of the above conditions but nevertheless have other features (e.g., learning difficulties, distinctive facial features) that suggest an underlying genetic cause. Nonspecific overgrowth is likely to be a heterogeneous group of conditions with multiple causes.

Management

No clinical practice guidelines for Sotos syndrome have been published. In the absence of published guidelines, the following recommendations are based on the authors' personal experience managing individuals with this disorder.

Evaluations Following Initial Diagnosis

To establish the extent of disease and needs in an individual diagnosed with Sotos syndrome, the evaluations summarized in Table 4 (if not performed as part of the evaluation that led to the diagnosis) are recommended [Tatton-Brown & Rahman 2007].

Table 4. Sotos Syndrome: Recommended Evaluations Following Initial Diagnosis

System/Concern	Evaluation	Comment
Constitutional	Assess serial growth measurements (stature, weight, & OFC).	
Development	Developmental assessment	- To incl motor, adaptive, cognitive, & speech-language eval - Eval for early intervention / special education

Table 4. continued from previous page.

System/Concern	Evaluation	Comment
Neurobehavioral/ Psychiatric	Neuropsychiatric eval	For persons age >12 mos: screening for concerns incl sleep disturbances, ADHD, anxiety, &/or findings suggestive of ASD
Cardiovascular	Cardiac eval	- To incl baseline echocardiogram - Blood pressure measurement
Genitourinary	Nephrology eval	- To include baseline renal US - In adults in whom diagnosis has just been established, renal US to evaluate for renal damage from quiescent chronic VUR - Assessment for cryptorchidism, hydrocele, hypospadias
Neurologic	Neurologic eval	- To incl brain MRI if progressive macrocephaly or unexplained neurologic features are present - Consider EEG if seizures are a concern.
Musculoskeletal	Orthopedics / physical medicine & rehab / PT & OT eval	To incl assessment of: - Gross motor & fine motor skills - Contractures & scoliosis - Need for PT (to improve gross motor skills) &/or OT (to improve fine motor skills)
Eyes	Ophthalmologic eval	To assess for reduced vision, abnormal ocular movement, best corrected visual acuity, refractive errors, strabismus, & more complex findings (e.g., cataract, retinal dystrophy) that may require referral for subspecialty care &/or low vision services
Gastrointestinal/ Feeding	Gastroenterology / nutrition / feeding team eval	To incl eval of GER, aspiration risk, & nutritional status
Audiologic	Audiology eval	To incl eval for hearing loss
Genetic counseling	By genetics professionals ¹	To obtain a pedigree & inform affected persons & their families re nature, MOI, & implications of Sotos syndrome to facilitate medical & personal decision making
Family support & resources	By clinicians, wider care team, & family support organizations	Assessment of family & social structure to determine need for: - Community or online resources such as Parent to Parent - Social work involvement for parental support - Home nursing referral

ADHD = attention-deficit hyperactivity disorder; ASD = autism spectrum disorder; GER = gastroesophageal reflux; MOI = mode of inheritance; OFC = orbitofrontal cortex; US = ultrasound; VUR = vesicoureteral reflux

1. Clinical geneticist, certified genetic counselor, certified genetic nurse, genetics advanced practice provider (nurse practitioner or physician assistant)

Treatment of Manifestations

There is no cure for Sotos syndrome. Supportive care to improve quality of life, maximize function, and reduce complications is recommended. This ideally involves multidisciplinary care by specialists in relevant fields (see Table 5).

Table 5. Sotos Syndrome: Treatment of Manifestations

Manifestation/Concern	Treatment	Considerations/Other
Developmental delay / Intellectual disability / Neurobehavioral issues	See Developmental Delay / Intellectual Disability Management Issues.	

Table 5. continued from previous page.

Manifestation/Concern	Treatment	Considerations/Other
Neurologic ¹	Standardized treatment w/ASM by experienced neurologist	- Many ASMs may be effective; none has been demonstrated effective specifically for this disorder. - Education of parents/caregivers²
Poor weight gain / Failure to thrive	Feeding therapy	Low threshold for clinical feeding eval &/or radiographic swallowing study when showing clinical signs or symptoms of dysphagia
Musculoskeletal	Orthopedics / physical medicine & rehab / PT & OT incl stretching to help avoid contractures & falls	Consider need for positioning & mobility devices, disability parking placard.
	Monitor for scoliosis	
Eyes	Ophthalmologist	Refractive errors, strabismus
	Ophthalmic subspecialist	More complex findings (e.g., cataract, retinal dystrophy)
Hearing	Hearing aids may be helpful per otolaryngologist.	Community hearing services through early intervention or school district
GI issues	Monitor for constipation & GER.	Stool softeners, prokinetics, osmotic agents, or laxatives as needed
Cardiac	Monitor for any cardiac defects.	
Genitourinary	Monitor for renal anomalies & VUR.	
Transition to adult care	Develop realistic plans for adult life (see American Epilepsy Society Transitions from Pediatric Epilepsy to Adult Epilepsy Care).	Starting by age ~10 yrs
Family/Community	- Ensure appropriate social work involvement to connect families w/local resources, respite, & support. - Coordinate care to manage multiple subspecialty appointments, equipment, medications, & supplies.	- Ongoing assessment of need for palliative care involvement &/or home nursing - Consider involvement in adaptive sports or Special Olympics.

ASM = anti-seizure medication; GER = gastroesophageal reflux; OT = occupational therapy; PT = physical therapy; VUR = vesicoureteral reflux

1. If brain MRI has been performed and ventricular dilatation demonstrated, shunting should not usually be necessary as the "arrested hydrocephalus" associated with Sotos syndrome is typically not obstructive and not associated with raised intracranial pressure. If raised intracranial pressure is suspected, investigation and management in consultation with neurologists and neurosurgeons would be appropriate.

2. Education of parents/caregivers regarding common seizure presentations is appropriate. For information on non-medical interventions and coping strategies for children diagnosed with epilepsy, see [Epilepsy Foundation Toolbox](#).

Developmental Delay / Intellectual Disability Management Issues

The following information represents typical management recommendations for individuals with developmental delay / intellectual disability in the United States; standard recommendations may vary from country to country.

Ages 0-3 years. Referral to an early intervention program is recommended for access to occupational, physical, speech, and feeding therapy as well as infant mental health services, special educators, and sensory impairment specialists. In the US, early intervention is a federally funded program available in all states that provides in-home services to target individual therapy needs.

Ages 3-5 years. In the US, developmental preschool through the local public school district is recommended. Before placement, an evaluation is made to determine needed services and therapies and an individualized education plan (IEP) is developed for those who qualify based on established motor, language, social, or cognitive delay. The early intervention program typically assists with this transition. Developmental preschool is center based; for children too medically unstable to attend, home-based services are provided.

All ages. Consultation with a developmental pediatrician is recommended to ensure the involvement of appropriate community, state, and educational agencies (US) and to support parents in maximizing quality of life. Some issues to consider:

- IEP services:
 - An IEP provides specially designed instruction and related services to children who qualify.
 - IEP services will be reviewed annually to determine whether any changes are needed.
 - Special education law requires that children participating in an IEP be in the least restrictive environment feasible at school and included in general education as much as possible, when and where appropriate.
 - Vision and hearing consultants should be a part of the child's IEP team to support access to academic material.
 - PT, OT, and speech services will be provided in the IEP to the extent that the need affects the child's access to academic material. Beyond that, private supportive therapies based on the affected individual's needs may be considered. Specific recommendations regarding type of therapy can be made by a developmental pediatrician.
 - As a child enters the teen years, a transition plan should be discussed and incorporated in the IEP. For those receiving IEP services, the public school district is required to provide services until age 21.
- A 504 plan (Section 504: a US federal statute that prohibits discrimination based on disability) can be considered for those who require accommodations or modifications such as front-of-class seating, assistive technology devices, classroom scribes, extra time between classes, modified assignments, and enlarged text.
- Developmental Disabilities Administration (DDA) enrollment is recommended. DDA is a US public agency that provides services and support to qualified individuals. Eligibility differs by state but is typically determined by diagnosis and/or associated cognitive/adaptive disabilities.
- Families with limited income and resources may also qualify for supplemental security income (SSI) for their child with a disability.

Motor Dysfunction

Gross motor dysfunction

- Physical therapy is recommended to maximize mobility and to reduce the risk for later-onset orthopedic complications (e.g., contractures, scoliosis, hip dislocation).
- Consider use of durable medical equipment and positioning devices as needed (e.g., wheelchairs, walkers, bath chairs, orthotics, adaptive strollers).
- For muscle tone abnormalities including hypertonia or dystonia, consider involving appropriate specialists to aid in management of baclofen, tizanidine, Botox®, anti-parkinsonian medications, or orthopedic procedures.

Fine motor dysfunction. Occupational therapy is recommended for difficulty with fine motor skills that affect adaptive function such as feeding, grooming, dressing, and writing.

Oral motor dysfunction should be assessed at each visit and clinical feeding evaluations and/or radiographic swallowing studies should be obtained for choking/gagging during feeds, poor weight gain, frequent respiratory

illnesses, or feeding refusal that is not otherwise explained. Assuming that the child is safe to eat by mouth, feeding therapy (typically from an occupational or speech therapist) is recommended to help improve coordination or sensory-related feeding issues. Feeds can be thickened or chilled for safety. When feeding dysfunction is severe, an NG-tube or G-tube may be necessary.

Communication issues. Consider evaluation for alternative means of communication (e.g., [augmentative and alternative communication](#) [AAC]) for individuals who have expressive language difficulties. An AAC evaluation can be completed by a speech-language pathologist who has expertise in the area. The evaluation will consider cognitive abilities and sensory impairments to determine the most appropriate form of communication. AAC devices can range from low-tech, such as picture exchange communication, to high-tech, such as voice-generating devices. Contrary to popular belief, AAC devices do not hinder verbal development of speech, but rather support optimal speech and language development.

Neurobehavioral/Psychiatric Concerns

Children may qualify for and benefit from interventions used in treatment of autism spectrum disorder, including applied behavior analysis (ABA). ABA therapy is targeted to the individual child's behavioral, social, and adaptive strengths and weaknesses and typically performed one on one with a board-certified behavior analyst.

Consultation with a developmental pediatrician may be helpful in guiding parents through appropriate behavior management strategies or providing prescription medications, such as medication used to treat attention-deficit/hyperactivity disorder, when necessary.

Concerns about serious aggressive or destructive behavior can be addressed by a pediatric psychiatrist.

Surveillance

To monitor existing manifestations, the individual's response to supportive care, and the emergence of new manifestations, the evaluations summarized in Table 6 are recommended [Tatton-Brown & Rahman 2007].

Table 6. Sotos Syndrome: Recommended Surveillance

System/Concern	Evaluation	Frequency
Feeding	- Measurement of growth parameters - Eval of nutritional status & safety of oral intake	At each visit
Gastrointestinal	Monitor for constipation.	
Neurologic	- Monitor those w/seizures as clinically indicated. - Assess for new manifestations such as seizures, changes in tone, & movement disorders.	
Development	Monitor developmental progress & educational needs.	
Neurobehavioral/ Psychiatric	Assessment for anxiety, ADHD, ASD, aggression, & self-injury	
Musculoskeletal	Physical medicine & OT/PT assessment of mobility & self-help skills	Per treating ophthalmologist(s)
Ophthalmologic involvement	- Monitor those w/eye abnormalities as clinically indicated. - Assess for changes in visual acuity.	
Cardiovascular	Monitor those w/cardiovascular abnormalities as clinically indicated.	Need for & timing of follow-up studies determined by cardiologist

Table 6. continued from previous page.

System/Concern	Evaluation	Frequency
Audiologic	Assess for signs & symptoms of hearing difficulties.	At each visit
Family/Community	Assess family need for social work support (e.g., palliative/respite care, home nursing, other local resources), care coordination, or follow-up genetic counseling if new questions arise (e.g., family planning).	

ADHD = attention-deficit/hyperactivity disorder; ASD = autism spectrum disorder; OT = occupational therapy; PT = physical therapy

Note: Cancer screening is not recommended [Villani et al 2017]; however, there should be a low threshold to investigate possible tumor-related symptoms.

- The absolute risk of sacrococcygeal teratoma and neuroblastoma is low (~1%) [Tatton-Brown et al 2005b, Tatton-Brown & Rahman 2007, Foster et al 2019]. This level of risk does not warrant routine screening, particularly as screening for neuroblastoma has not been shown to decrease mortality and can lead to false positive results [Schilling et al 2002].
- Wilms tumor risk is not significantly increased and routine renal ultrasound examination is not indicated [Scott et al 2006].

Evaluation of Relatives at Risk

See Genetic Counseling for issues related to testing of at-risk relatives for genetic counseling purposes.

Therapies Under Investigation

Search [ClinicalTrials.gov](https://clinicaltrials.gov) in the US and [EU Clinical Trials Register](https://clinicaltrialsregister.eu) in Europe for information on clinical studies for a wide range of diseases and conditions. Note: There may not be clinical trials for this disorder.

Genetic Counseling

Genetic counseling is the process of providing individuals and families with information on the nature, mode(s) of inheritance, and implications of genetic disorders to help them make informed medical and personal decisions. The following section deals with genetic risk assessment and the use of family history and genetic testing to clarify genetic status for family members; it is not meant to address all personal, cultural, or ethical issues that may arise or to substitute for consultation with a genetics professional. —ED.

Mode of Inheritance

Sotos syndrome is inherited in an autosomal dominant manner.

Risk to Family Members

Parents of a proband

- About 5% of individuals diagnosed with Sotos syndrome have an affected parent.
- Approximately 95% of individuals diagnosed with Sotos syndrome have the disorder as the result of a *de novo* genetic alteration (i.e., an *NSD1* pathogenic variant or a deletion encompassing *NSD1*).
- If neither parent of an individual with a molecular diagnosis of Sotos syndrome has clinical features of Sotos syndrome, it is very unlikely a parent is heterozygous for a Sotos syndrome-related genetic alteration. Molecular genetic testing of the parent for the genetic alteration identified in the proband can be used to confirm the genetic status of the parents and inform recurrence risk assessment.

- If the Sotos syndrome-related genetic alteration identified in the proband is not identified in either parent and parental identity testing has confirmed biological maternity and paternity, the following possibilities should be considered:
 - The proband has a *de novo* pathogenic variant.
 - The proband inherited a pathogenic variant from a parent with gonadal (or somatic and gonadal) mosaicism [Kamien et al 2016, Hai et al 2023]. Note: Testing of parental leukocyte DNA may not detect all instances of somatic mosaicism and will not detect a pathogenic variant that is present in the germ (gonadal) cells only.

Sibs of a proband. The risk to the sibs of a proband depends on the clinical/genetic status of the proband's parents:

- If a parent of the proband is affected and/or has an *NSD1* pathogenic variant or a deletion encompassing *NSD1*, the risk to the sibs of inheriting the genetic alteration is 50%.
- Intrafamilial clinical variability has been reported; sibs with the same Sotos syndrome-related genetic alteration can be affected differently [Tatton-Brown et al 2005b, Donnelly et al 2011].
- If the genetic status of the parents is unknown and both parents are clinically unaffected, the risk to the sibs of a proband appears to be low. However, sibs of a proband with clinically unaffected parents are still presumed to be at increased risk for Sotos syndrome because of the possibility of parental gonadal mosaicism. Parental somatic/gonadal mosaicism has been reported in the unaffected father of an affected individual [Kamien et al 2016] and presumed in the unaffected mother of a child with Sotos syndrome and a fetus with the same *NSD1* pathogenic variant [Hai et al 2023].

Offspring of a proband

- Each child of an individual with Sotos syndrome has a 50% chance of inheriting the *NSD1* pathogenic variant or deletion encompassing *NSD1*.
- Phenotypic expression can vary from one generation to the next; thus, it is not possible to accurately predict how heterozygous offspring may be affected.

Other family members. The risk to other family members depends on the status of the proband's parents: if a parent has the Sotos syndrome-related genetic alteration, the parent's family members may be at risk.

Related Genetic Counseling Issues

Family planning

- The optimal time for determination of genetic risk and discussion of the availability of prenatal/preimplantation genetic testing is before pregnancy.
- It is appropriate to offer genetic counseling (including discussion of potential risks to offspring and reproductive options) to young adults who are affected.

Prenatal Testing and Preimplantation Genetic Testing

Molecular genetic testing. Once the Sotos syndrome-related genetic alteration has been identified in an affected family member, prenatal and preimplantation genetic testing are possible. Note that phenotypic expression can vary from one generation to the next; thus, it is not possible to accurately predict phenotype based on the prenatal finding of a Sotos syndrome-related genetic alteration.

Differences in perspective may exist among medical professionals and within families regarding the use of prenatal and preimplantation genetic testing. While most health care professionals would consider use of prenatal and preimplantation genetic testing to be a personal decision, discussion of these issues may be helpful.

Ultrasound examination. Prenatal diagnosis cannot be accurately accomplished by ultrasound examination: the features of Sotos syndrome likely to be detected by ultrasound examination, such as macrocephaly and increased length, are nonspecific.

Resources

GeneReviews staff has selected the following disease-specific and/or umbrella support organizations and/or registries for the benefit of individuals with this disorder and their families. GeneReviews is not responsible for the information provided by other organizations. For information on selection criteria, click [here](#).

- **Sotos Syndrome Support Association (SSSA)**

Phone: 888-246-7772

Email: info@sotosyndrome.org
sotosyndrome.org

- **Child Growth Foundation**

United Kingdom

Phone: 0208 995 0257

Email: info@childgrowthfoundation.org
childgrowthfoundation.org

Molecular Genetics

Information in the Molecular Genetics and OMIM tables may differ from that elsewhere in the GeneReview: tables may contain more recent information. —ED.

Table A. Sotos Syndrome: Genes and Databases

Gene	Chromosome Locus	Protein	Locus-Specific Databases	HGMD	ClinVar
NSD1	5q35.3	Histone-lysine N-methyltransferase, H3 lysine-36 specific	Nuclear receptor binding SET Domain protein 1 (NSD1) @ LOVD	NSD1	NSD1

Data are compiled from the following standard references: gene from [HGNC](#); chromosome locus from [OMIM](#); protein from [UniProt](#). For a description of databases (Locus Specific, HGMD, ClinVar) to which links are provided, click [here](#).

Table B. OMIM Entries for Sotos Syndrome ([View All in OMIM](#))

117550	SOTOS SYNDROME; SOTOS
606681	NUCLEAR RECEPTOR-BINDING SET DOMAIN PROTEIN 1; NSD1

NSD1 encodes histone-lysine N-methyltransferase, H3 lysine-36 specific (also called nuclear receptor-binding SET domain-containing protein 1; NSD1) and is expressed in the brain, kidney, skeletal muscle, spleen, thymus, and lung. NSD1 contains at least 12 functional domains:

- Two nuclear receptor interaction domains (NID^{-L} and NID^{+L})
- Two proline-tryptophan-tryptophan-proline (PWWP) domains
- Five plant homeo domains (PHD)
- A SET (su(var)3-9, enhancer of zeste, trithorax) domain

The most distinctive of these domains are the SET and associated SAC (SET-associated Cys-rich) domains, which are found in histone methyltransferases that regulate chromatin states. The SET domain of NSD1 has unique histone specificity, methylating lysine residue 36 on histone H3, and lysine residue 20 on histone H4

(K36H3 and K20H4) [Rayasam et al 2003]. PHDs are also typically found in proteins that act at the chromatin level, and PWWP domains are implicated in protein-protein interactions and are often found in methyltransferases. The nuclear receptors of NSD1, NID^{-L}, and NID^{+L} are typical of those found in corepressors and coactivators [Huang et al 1998]. The presence of these distinctive domains suggests that NSD1 is a histone methyltransferase that acts as a transcriptional intermediary factor capable of both negatively and positively influencing transcription, depending on the cellular context [Kurotaki et al 2001].

Mechanism of disease causation. Loss of function. NSD1 is a histone methyltransferase. Variants associated with Sotos syndrome result in loss of NSD1 function [Douglas et al 2003]. It is thought that disrupted histone methyltransferase activity, which plays an important role in the expression of growth regulatory genes in addition to broader epigenetic changes, results in the phenotypic features of Sotos syndrome. However, further study is needed to determine the precise mechanism of disease causation.

Chapter Notes

Revision History

- 5 June 2025 (gm) Comprehensive update posted live
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- 8 March 2012 (me) Comprehensive update posted live
- 10 December 2009 (me) Comprehensive update posted live
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- 26 May 2004 (tc) Original submission

References

Literature Cited

- Agwu JC, Shaw NJ, Kirk J, Chapman S, Ravine D, Cole TR. Growth in Sotos syndrome. *Arch Dis Child*. 1999;80:339-42 PubMed PMID: 10086939.
- Allanson JE, Cole TR. Sotos syndrome: evolution of facial phenotype subjective and objective assessment. *Am J Med Genet*. 1996;65:13-20. PubMed PMID: 8914735.
- Aref-Eshghi E, Bend EG, Colaiacovo S, Caudle M, Chakrabarti R, Napier M, Brick L, Brady L, Carere DA, Levy MA, Kerkhof J, Stuart A, Saleh M, Beaudet AL, Li C, Kozenko M, Karp N, Prasad C, Siu VM, Tarnopolsky MA, Ainsworth PJ, Lin H, Rodenhiser DI, Krantz ID, Deardorff MA, Schwartz CE, Sadikovic B. Diagnostic utility of genome-wide DNA methylation testing in genetically unsolved individuals with suspected hereditary conditions. *Am J Hum Genet*. 2019;104:685–700. PubMed PMID: 30929737.
- Aref-Eshghi E, Kerkhof J, Pedro VP, Groupe DI, France, Barat-Houari M, Ruiz-Pallares N, Andrau JC, Lacombe D, Van-Gils J, Fergelot P, Dubourg C, Cormier-Daire V, Rondeau S, Lecoquierre F, Saugier-Verber P, Nicolas G, Lesca G, Chatron N, Sanlaville D, Vitobello A, Faivre L, Thauvin-Robinet C, Laumonnier F, Raynaud M, Alders M, Mannens M, Henneman P, Hennekam RC, Velasco G, Francastel C, Ulveling D, Ciolfi A, Pizzi S, Tartaglia M, Heide S, Héron D, Mignot C, Keren B, Whalen S, Afenjar A, Bienvenu T, Campeau PM, Rousseau J, Levy MA, Brick L, Kozenko M, Balci TB, Siu VM, Stuart A, Kadour M, Masters J, Takano K, Kleefstra T, de Leeuw N, Field M, Shaw M, Gecz J, Ainsworth PJ, Lin H, Rodenhiser DI, Friez MJ, Tedder M, Lee JA, DuPont BR, Stevenson RE, Skinner SA, Schwartz CE, Genevieve D, Sadikovic B. Evaluation of DNA methylation epigenatures for diagnosis and phenotype correlations in 42 mendelian neurodevelopmental disorders. *Am J Hum Genet*. 2020;106:356–70. PubMed PMID: 32109418.

- Buxbaum JD, Cai G, Nygren G, Chaste P, Delorme R, Goldsmith J, Råstam M, Silverman JM, Hollander E, Gillberg C, Leboyer M, Betancur C. Mutation analysis of the NSD1 gene in patients with autism spectrum disorders and macrocephaly. *BMC Med Genet*. 2007;8:68 PubMed PMID: 18001468.
- Calcagni G, Ferrigno F, Franceschini A, Dentici ML, Capolino R, Sinibaldi L, Minotti C, Micalizzi A, Alesi V, Novelli A, Baban A. Congenital heart defects in patients with molecularly confirmed Sotos syndrome. *Diagnostics*. 2024;14:594. PubMed PMID: 38535015.
- Choi N, Kim HY, Ko JM. Deciphering growth patterns in Korean children with Sotos syndrome through the development of a disease-specific growth chart. *Mol Genet Genomic Med*. 2024;12:e70028. PubMed PMID: 39494594.
- Cole TR, Hughes HE. Sotos syndrome: a study of the diagnostic criteria and natural history. *J Med Genet*. 1994;31:20-32. PubMed PMID: 7512144.
- Deardorff MA, Maisenbacher M, Zackai EH. Ganglioglioma in a Sotos syndrome patient with an NSD1 deletion. *Am J Med Genet A*. 2004;130A:393-4. PubMed PMID: 15455365.
- de Boer L, le Cessie S, Wit JM. Auxological data in patients clinically suspected of Sotos syndrome with NSD1 gene alterations. *Acta Paediatr*. 2005;94:1142-4 PubMed PMID: 16188863.
- de Boer L, Roder I, Wit JM. Psychosocial, cognitive, and motor functioning in patients with suspected Sotos syndrome: a comparison between patients with and without NSD1 gene alterations. *Dev Med Child Neurol*. 2006;48:582-8 PubMed PMID: 16780628.
- Donnelly DE, Turnpenny P, McConnell VP. Phenotypic variability in a three-generation Northern Irish family with Sotos syndrome. *Clin Dysmorphol*. 2011;20:175-81 PubMed PMID: 21738022.
- Douglas J, Hanks S, Temple IK, Davies S, Murray A, Upadhyaya M, Tomkins S, Hughes HE, Cole TR, Rahman N. NSD1 mutations are the major cause of Sotos syndrome and occur in some cases of Weaver syndrome but are rare in other overgrowth phenotypes. *Am J Hum Genet*. 2003;72:132-43 PubMed PMID: 12464997.
- Faivre L, Viot G, Prieur M, Turleau C, Gosset P, Romana S, Munnich A, Vekemans M, Cormier-Daire V. Apparent Sotos syndrome (cerebral gigantism) in a child with trisomy 20p11.2-p12.1 mosaicism. *Am J Med Genet*. 2000;91:273-6 PubMed PMID: 10766982.
- Finegan JK, Cole TR, Kingwell E, Smith ML, Smith M, Sitarenios G. Language and behavior in children with Sotos syndrome. *J Am Acad Child Adolesc Psychiatry*. 1994;33:1307-15 PubMed PMID: 7995798.
- Foster A, Zachariou A, Loveday C, Ashraf T, Blair E, Clayton-Smith J, Dorkins H, Fryer A, Gener B, Goudie D, Henderson A, Irving M, Joss S, Keeley V, Lahiri N, Lynch SA, Mansour S, McCann E, Morton J, Motton N, Murray A, Riches K, Shears D, Stark Z, Thompson E, Vogt J, Wright M, Cole T, Tatton-Brown K. The phenotype of Sotos syndrome in adulthood: a review of 44 individuals. *Am J Med Genet C Semin Med Genet*. 2019;181:502-8. PubMed PMID: 31479583.
- Frattale I, Sarnataro R, Siracusano M, Riccioni A, Galasso C, Valeriani M, Conteduca G, Coviello D, Mazzone L, Moavero R. Sleep disturbances and behavioral symptoms in pediatric Sotos syndrome. *Front Neurol*. 2024;15:1360055. PubMed PMID: 38434199.
- Grand K, Gonzalez-Gandolfi C, Ackermann AM, Aljeaid D, Bedoukian E, Bird LM, De Leon DD, Diaz J, Hopkin RJ, Kadakia SP, Keena B. Hyperinsulinemic hypoglycemia in seven patients with de novo NSD1 mutations. *Am J Med Genet A*. 2019;179:542-51. PubMed PMID: 30719864.
- Hai X, Liangjie G, Qian Z, Tao L, Hongdan W, Bing K, Ke Y, Qiannan G, Dong W, Shixiu L. Genetic analysis of a case of Sotos syndrome with suspected germinal mosaicism in mother. *Appl Biochem Biotechnol*. 2023;195:5792-801. PubMed PMID: 36708490.
- Hersh JH, Cole TR, Bloom AS, Bertolone SJ, Hughes HE. Risk of malignancy in Sotos syndrome. *J Pediatr*. 1992;120:572-4 PubMed PMID: 1552397.

- Hood RL, McGillivray G, Hunter MF, Roberston SP, Bulman DE, Boycott KM, Stark Z, et al. Severe connective tissue laxity including aortic dilatation in Sotos syndrome. *Am J Med Genet A*. 2016;170A:531-5 PubMed PMID: 26613968.
- Huang N, vom Baur E, Garnier JM, Lerouge T, Vonesch JL, Lutz Y, Chambon P, Losson R. Two distinct nuclear receptor interaction domains in NSD1, a novel SET protein that exhibits characteristics of both corepressors and coactivators. *EMBO J*. 1998;17:3398-412 PubMed PMID: 9628876.
- Husson T, Lecoquierre F, Nicolas G, Richard AC, Afenjar A, Audebert-Bellanger S, Badens C, Bilan F, Bizaoui V, Boland A, Bonnet-Dupeyron MN. Episignatures in practice: independent evaluation of published episignatures for the molecular diagnostics of ten neurodevelopmental disorders. *Eur J Hum Genet*. 2024;32:190-9. PubMed PMID: 37872275.
- Kamien B, Dadd T, Buckman M, Ronan A, Dudding T, Meldrum C, Scott R, Mina K. Somatic-gonadal mosaicism causing Sotos syndrome. *Am J Med Genet A*. 2016;170:3360-2 PubMed PMID: 27604501.
- Kurotaki N, Harada N, Shimokawa O, Miyake N, Kawame H, Uetake K, Makita Y, Kondoh T, Ogata T, Hasegawa T, Nagai T, Ozaki T, Touyama M, Shenhav R, Ohashi H, Medne L, Shiihara T, Ohtsu S, Kato Z, Okamoto N, Nishimoto J, Lev D, Miyoshi Y, Ishikiriya S, Sonoda T, Sakazume S, Fukushima Y, Kurosawa K, Cheng JF, Yoshiura K, Ohta T, Kishino T, Niikawa N, Matsumoto N. Fifty microdeletions among 112 cases of Sotos syndrome: low copy repeats possibly mediate the common deletion. *Hum Mutat*. 2003;22:378-87 PubMed PMID: 14517949.
- Kurotaki N, Harada N, Yoshiura K, Sugano S, Niikawa N, Matsumoto N. Molecular characterization of NSD1, a human homologue of the mouse *Nsd1* gene. *Gene*. 2001;279:197-204 PubMed PMID: 11733144.
- Lane C, Milne E, Freeth M. Characteristics of autism spectrum disorder in Sotos syndrome. *J Autism Dev Disord*. 2017;47:135-43 PubMed PMID: 27771801.
- Lane C, Milne E, Freeth M. The cognitive profile of Sotos syndrome. *J Neuropsychol*. 2019;13:240-52. PubMed PMID: 29336120.
- Levy MA, McConkey H, Kerkhof J, Barat-Houari M, Bargiacchi S, Biamino E, Bralo MP, Cappuccio G, Ciolfi A, Clarke A, DuPont BR, Elting MW, Faivre L, Fee T, Fletcher RS, Cherik F, Foroutan A, Friez MJ, Gervasini C, Haghsheenas S, Hilton BA, Jenkins Z, Kaur S, Lewis S, Louie RJ, Maitz S, Milani D, Morgan AT, Oegema R, Østergaard E, Pallares NR, Piccione M, Pizzi S, Plomp AS, Poulton C, Reilly J, Relator R, Rius R, Robertson S, Rooney K, Rousseau J, Santen GWE, Santos-Simarro F, Schijns J, Squeo GM, St John M, Thauvin-Robinet C, Traficante G, van der Sluijs PJ, Vergano SA, Vos N, Walden KK, Azmanov D, Balci T, Banka S, Gecz J, Henneman P, Lee JA, Mannens MMAM, Roscioli T, Siu V, Amor DJ, Baynam G, Bend EG, Boycott K, Brunetti-Pierri N, Campeau PM, Christodoulou J, Dymont D, Esber N, Fahrner JA, Fleming MD, Genevieve D, Kerrnohan KD, McNeill A, Menke LA, Merla G, Prontera P, Rockman-Greenberg C, Schwartz C, Skinner SA, Stevenson RE, Vitobello A, Tartaglia M, Alders M, Tedder ML, Sadikovic B. Novel diagnostic DNA methylation episignatures expand and refine the epigenetic landscapes of Mendelian disorders. *HGG Adv*. 2021;3:100075. PubMed PMID: 35047860.
- Martinez HR, Belmont JW, Craigen WJ, Taylor MD, Jefferies JL. Left ventricular noncompaction in Sotos syndrome. *Am J Med Genet A*. 2011;155A:1115-8. PubMed PMID: 21484993.
- Minatogawa M, Tsuji T, Inaba M, Kawakami N, Mizuno S, Kosho T. Atypical Sotos syndrome caused by a novel splice site variant. *Human Genome Variation*. 2022;9:41. PubMed PMID: 36379925.
- Nakamura A, Horigome H, Ishizu T, Seo Y, Sumazaki R. Double orifice mitral valve combined with left ventricular noncompaction in a child with Sotos syndrome. *J Cardiol Cases*. 2014;11:88-90. PubMed PMID: 30546538.
- Pezzani L, Mauri L, Selicorni A, Peron A, Grasso M, Codazzi AC, Rimini A, Marchisio PG, Coviello D, Colli A, Milani D. Aortic dilation in Sotos syndrome: an underestimated feature. *Am J Med Genet A*. 2020;182:1819-23. PubMed PMID: 32286744.

- Rayasam GV, Wendling O, Angrand PO, Mark M, Niederreither K, Song L, Lerouge T, Hager GL, Chambon P, Losson R. NSD1 is essential for early post-implantation development and has a catalytically active SET domain. *EMBO J*. 2003;22:3153-63 PubMed PMID: 12805229.
- Richards S, Aziz N, Bale S, Bick D, Das S, Gastier-Foster J, Grody WW, Hegde M, Lyon E, Spector E, Voelkerding K, Rehm HL, et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genet Med*. 2015;17:405-24. PubMed PMID: 25741868.
- Rio M, Clech L, Amiel J, Faivre L, Lyonnet S, Le Merrer M, Odent S, Lacombe D, Edery P, Brauner R, Raoul O, Gosset P, Prieur M, Vekemans M, Munnich A, Colleaux L, Cormier-Daire V. Spectrum of NSD1 mutations in Sotos and Weaver syndromes. *J Med Genet*. 2003;40:436-40 PubMed PMID: 12807965.
- Saccucci P, Papetti F, Martinoli R, Dofcaci A, Tuderti U, Marcantonio A, Di Renzi P, Fahim A, Ferrante F, Banci M. Isolated left ventricular noncompaction in a case of Sotos syndrome: a casual or causal link? *Cardiol Res Pract*. 2011;2011:824095. PubMed PMID: 21747990.
- Saugier-Verber P, Bonnet C, Afejar A, Drouin-Garraud V, Coubes C, Fehrenbach S, Holder-Espinasse M, Roume J, Malan V, Portnoi MF, Jeanne N. Heterogeneity of NSD1 alterations in 116 patients with Sotos syndrome. *Hum Mutat*. 2007;28:1098-107. PubMed PMID: 17565729.
- Schilling FH, Spix C, Berthold F, Erttmann R, Fehse N, Hero B, Klein G, Sander J, Schwarz K, Treuner J, Zorn U, Michaelis J. Neuroblastoma screening at one year of age. *N Engl J Med*. 2002;346:1047-53 PubMed PMID: 11932471.
- Scott RH, Walker L, Olsen ØE, Levitt G, Kenney I, Maher E, Owens CM, Pritchard-Jones K, Craft A, Rahman N. Surveillance for Wilms tumour in at-risk children: pragmatic recommendations for best practice. *Arch Dis Child*. 2006;91:995-9 PubMed PMID: 16857697.
- Sheth K, Moss J, Hyland S, Stinton C, Cole T, Oliver C. The behavioral characteristics of Sotos syndrome. *Am J Med Genet A*. 2015;167A:2945-56 PubMed PMID: 26418839.
- Stenson PD, Mort M, Ball EV, Chapman M, Evans K, Azevedo L, Hayden M, Heywood S, Millar DS, Phillips AD, Cooper DN. The Human Gene Mutation Database (HGMD®): optimizing its use in a clinical diagnostic or research setting. *Hum Genet*. 2020;139:1197-207. PubMed PMID: 32596782.
- Takkar A, Arora V, Saxena R, Kumar P, Verma IC. Reversal of clinical phenotype of Sotos syndrome due to microduplication of NSD1 gene. *Indian J Pediatr*. 2022;89:1137-39. PubMed PMID: 35925544.
- Tatton-Brown K, Douglas J, Coleman K, Baujat G, Chandler K, Clarke A, Collins A, Davies S, Faravelli F, Firth H, Garrett C, Hughes H, Kerr B, Liebelt J, Reardon W, Schaefer GB, Splitt M, Temple IK, Waggoner D, Weaver DD, Wilson L, Cole T, Cormier-Daire V, Irrthum A, Rahman N. Multiple mechanisms are implicated in the generation of 5q35 microdeletions in Sotos syndrome. *J Med Genet*. 2005a;42:307-13 PubMed PMID: 15805156.
- Tatton-Brown K, Douglas J, Coleman K, Baujat G, Cole TR, Das S, Horn D, Hughes HE, Temple IK, Faravelli F, Waggoner D, Turkmen S, Cormier-Daire V, Irrthum A, Rahman N. Genotype-phenotype associations in Sotos syndrome: an analysis of 266 individuals with NSD1 aberrations. *Am J Hum Genet*. 2005b;77:193-204 PubMed PMID: 15942875.
- Tatton-Brown K, Loveday C, Yost S, Clarke M, Ramsay E, Zachariou A, Elliott A, Wylie H, Ardisson A, Rittinger O, Stewart F. Mutations in epigenetic regulation genes are a major cause of overgrowth with intellectual disability. *Am J Hum Genet*. 2017;100:725-36. PubMed PMID: 28475857.
- Tatton-Brown K, Rahman N. Clinical features of NSD1-positive Sotos syndrome. *Clin Dysmorphol*. 2004;13:199-204 PubMed PMID: 15365454.
- Tatton-Brown K, Rahman N. Sotos syndrome. *Eur J Hum Genet*. 2007;15:264-71 PubMed PMID: 16969376.

- Testa B, Conteduca G, Grasso M, Cecconi M, Lantieri F, Baldo C, Arado A, Andraghetti L, Malacarne M, Milani D, Coviello D. Molecular analysis and reclassification of NSD1 gene variants in a cohort of patients with clinical suspicion of Sotos syndrome. *Genes*. 2023;14:295. PubMed PMID: 36833222.
- Theodoulou E, Baborie A, Jenkinson MD. Low grade glioma in an adult patient with Sotos syndrome. *J Clin Neurosci*. 2015;22:413-5 PubMed PMID: 25304431.
- Türkmen S, Gillissen-Kaesbach G, Meinecke P, Albrecht B, Neumann LM, Hesse V, Palanduz S, Balg S, Majewski F, Fuchs S, Zscheschang P, Greiwe M, Mennicke K, Kreuz FR, Dehmel HJ, Rodeck B, Kunze J, Tinschert S, Mundlos S, Horn D. Mutations in NSD1 are responsible for Sotos syndrome, but are not a frequent finding in other overgrowth phenotypes. *Eur J Hum Genet*. 2003;11:858-65. PubMed PMID: 14571271.
- van der Lugt NM, Weerts MJA, Veenma DCM, Lincke CR, Gischler SJ, Alders M, van Ierland Y. 5q35 duplication syndrome: Narrowing the critical region on the distal side and further evidence of intrafamilial variability and expression. *Am J Med Genet A*. 2023;191:835-41. PubMed PMID: 36458506.
- Villani A, Greer MC, Kalish JM, Nakagawara A, Nathanson KL, Pajtler KW, Pfister SM, Walsh MF, Wasserman JD, Zelley K, Kratz CP. Recommendations for cancer surveillance in individuals with RASopathies and other rare genetic conditions with increased cancer risk. *Clin Cancer Res*. 2017;23:e83-e90. PubMed PMID: 28620009.
- Villate O, Maortua H, Tejada MI, Llano-Rivas I. RNA Analysis and clinical characterization of a novel splice variant in the NSD1 gene causing familial sotos syndrome. *Front Pediatr*. 2022;10:827802. PubMed PMID: 35186810.
- Visser R, Shimokawa O, Harada N, Niikawa N, Matsumoto N. Non-hotspot-related breakpoints of common deletions in Sotos syndrome are located within destabilised DNA regions. *J Med Genet*. 2005;42:e66. PubMed PMID: 16272258.
- Waggoner DJ, Raca G, Welch K, Dempsey M, Anderes E, Ostrovnya I, Alkhateeb A, Kamimura J, Matsumoto N, Schaeffer GB, Martin CL, Das S. NSD1 analysis for Sotos syndrome: insights and perspectives from the clinical laboratory. *Genet Med*. 2005;7:524-33 PubMed PMID: 16247291.
- Yue X, Liu B, Han T, Guo D, Ding R, Wang G. The first pineoblastoma case report of a patient with Sotos syndrome harboring NSD1 germline mutation. *BMC pediatrics*. 2024;24:166. PubMed PMID: 38459438.

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